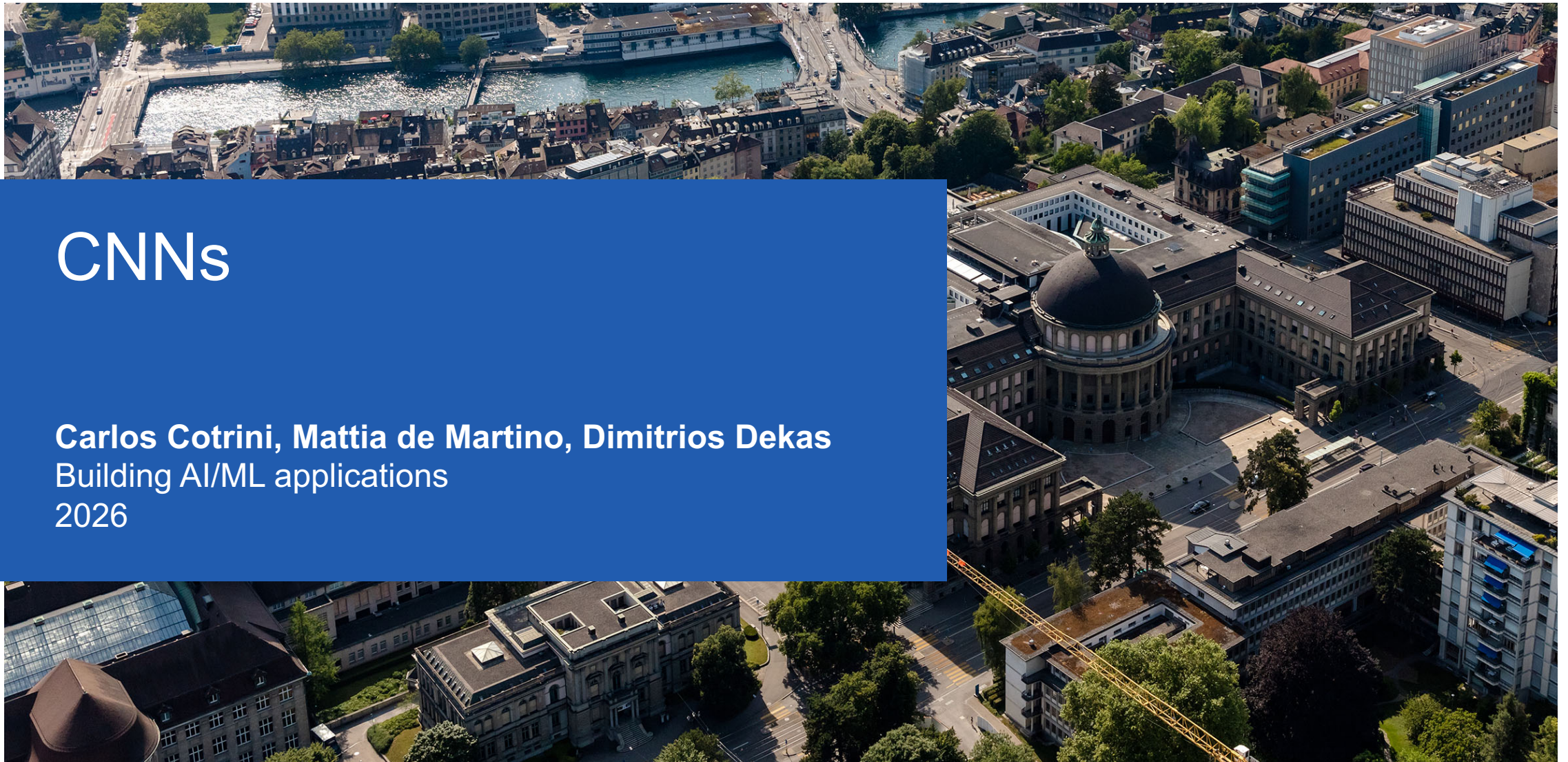


CNNs

Carlos Cotrini, Mattia de Martino, Dimitrios Dekas
Building AI/ML applications
2026



Four main tasks in computer vision

Regression and
classification of images



Cat? Yes

Dog? Yes

Bat? No

Feature extraction



Cat color: black

Dog color: brown

Cat's eye color:
green

Dog's eye color:
brown

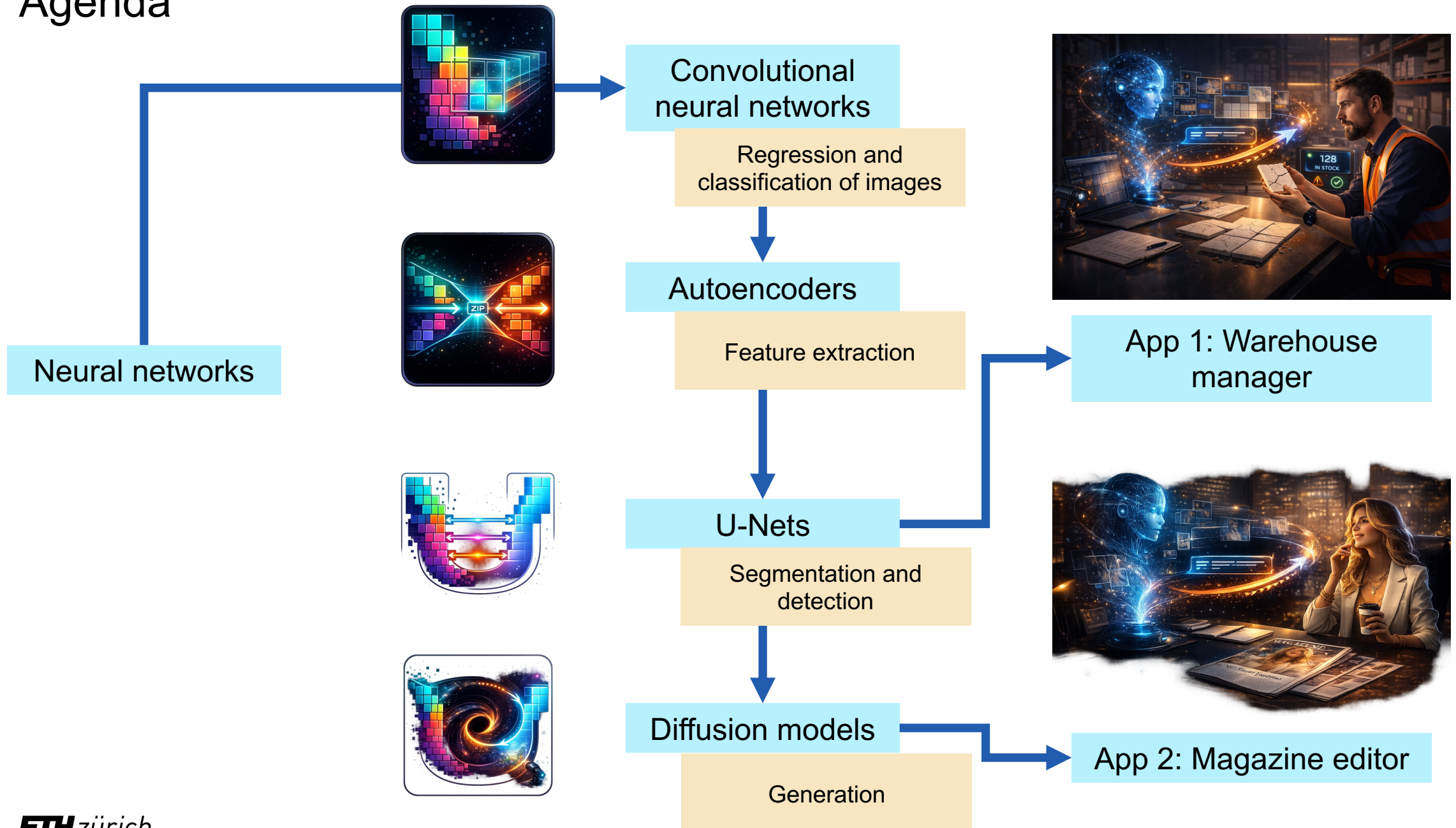
Segmentation and
detection



Generation



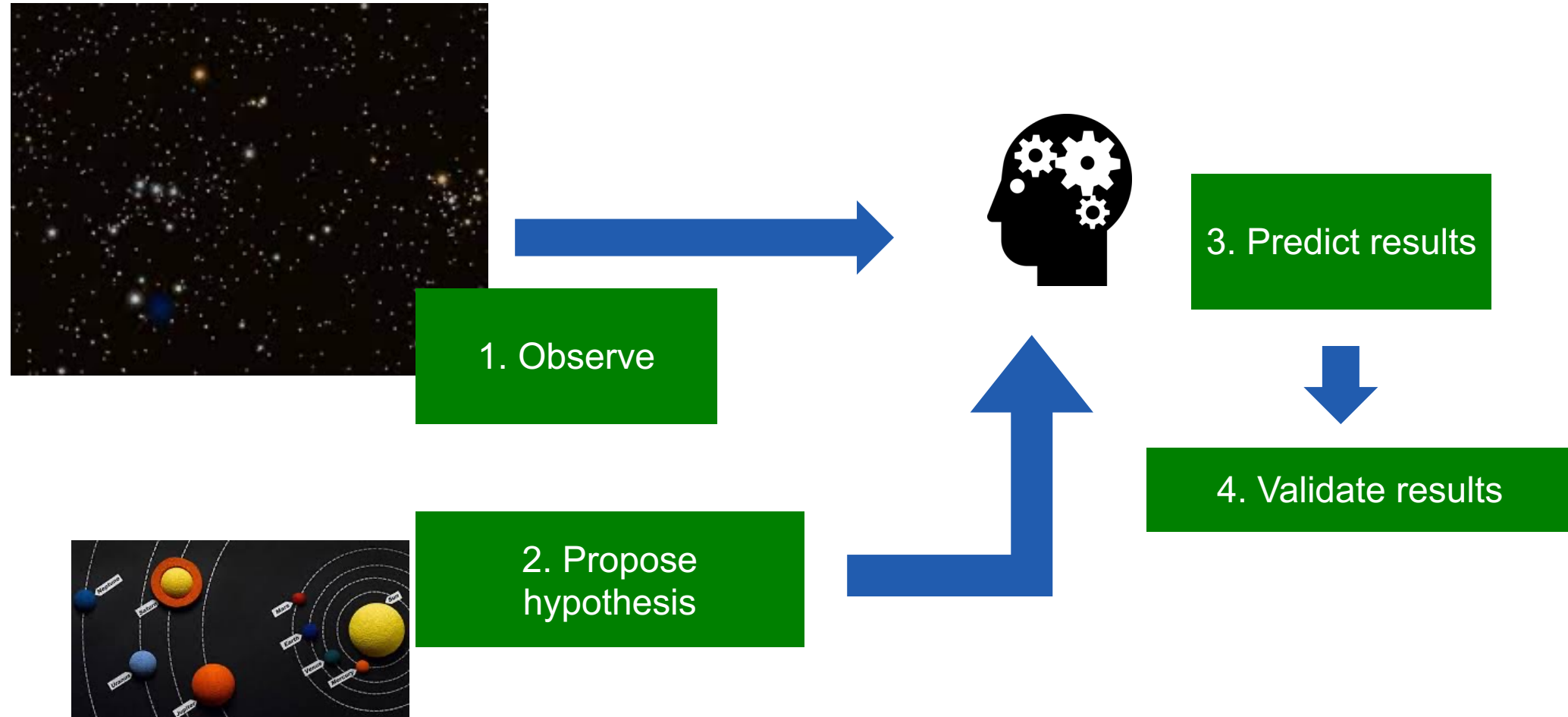
Agenda



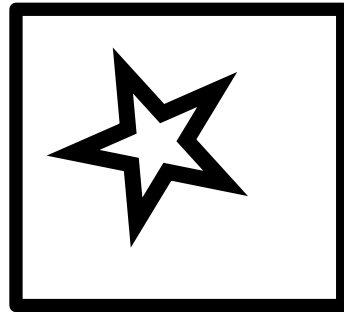
Motivation

What is learning?

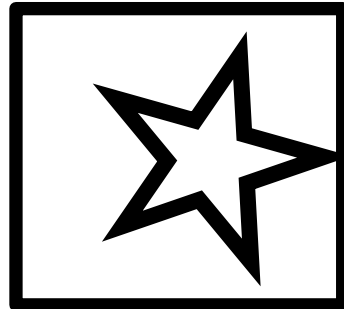
How humans create knowledge



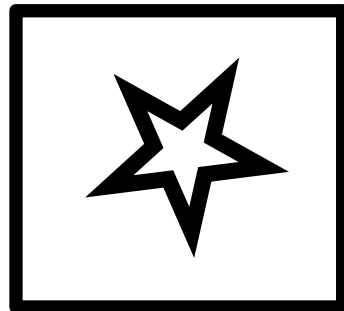
Classifying stars and planets



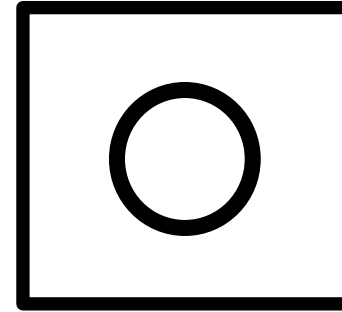
Star



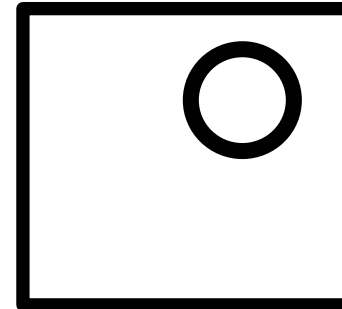
Star



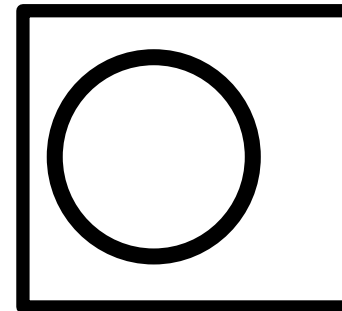
Star



Planet

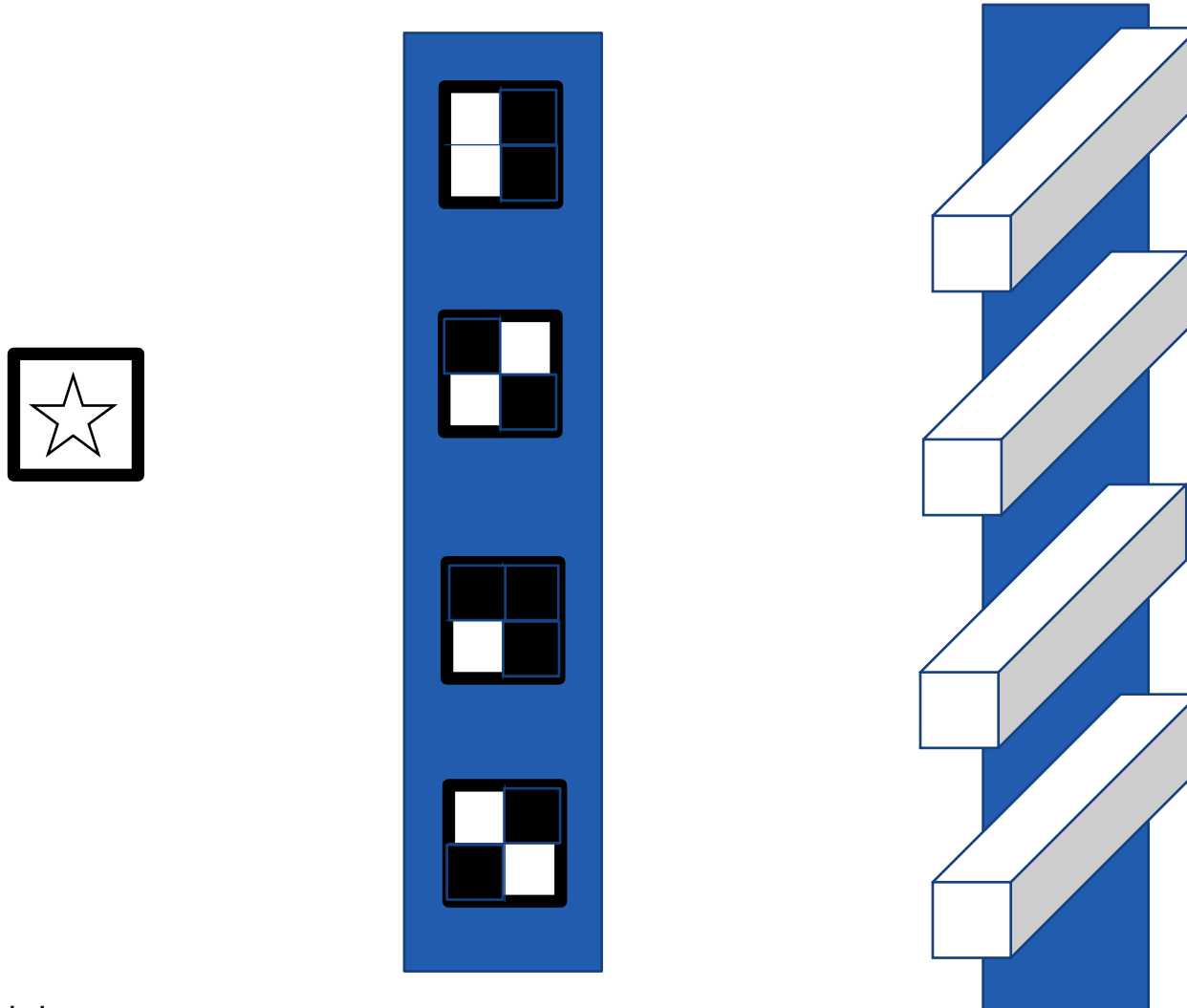


Planet

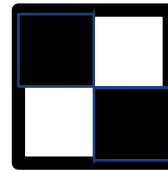
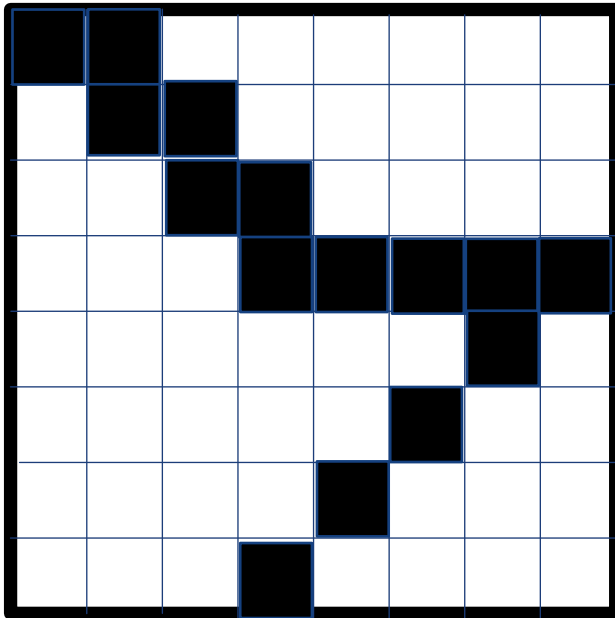


Planet

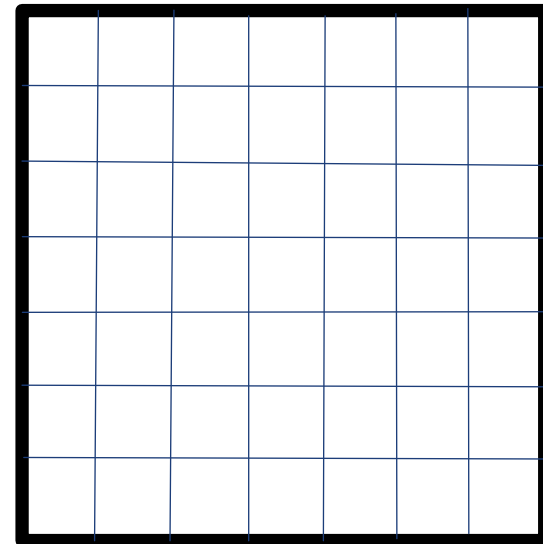
Convolutional neural networks



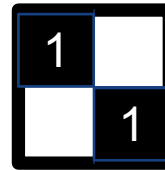
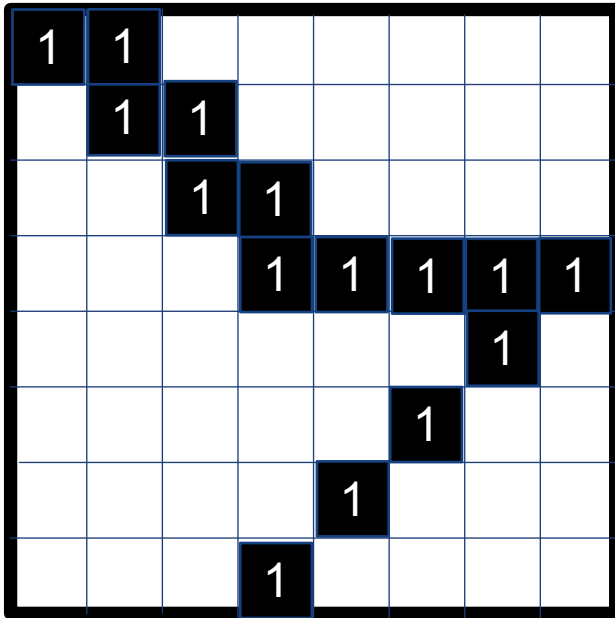
Convolutional neural networks



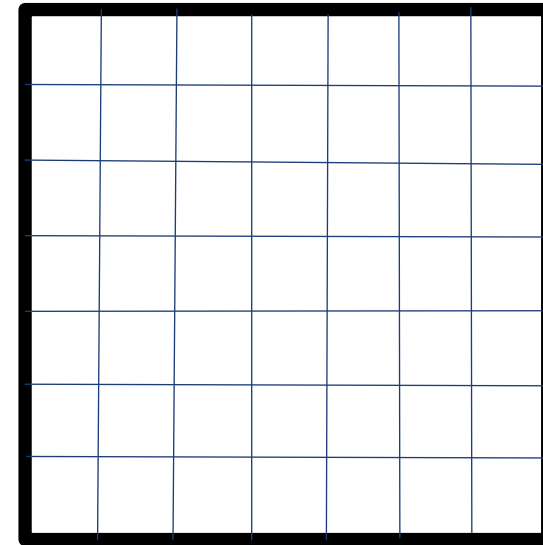
Convolutional filter



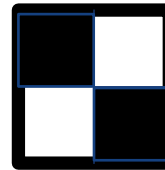
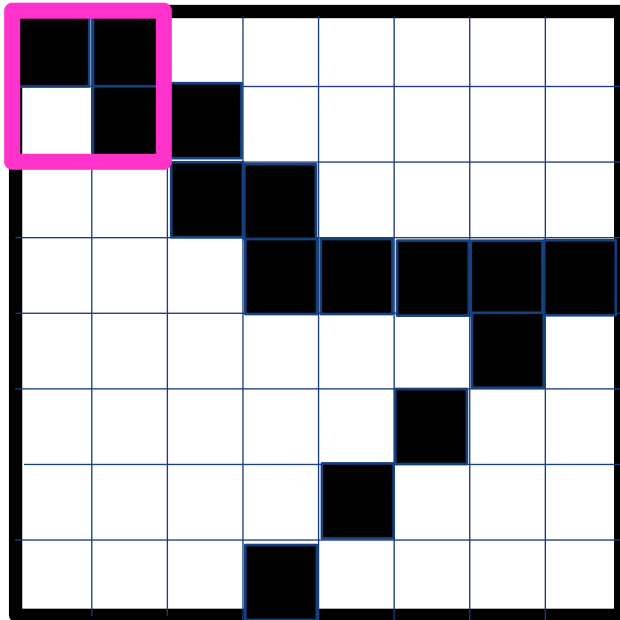
Convolutional neural networks



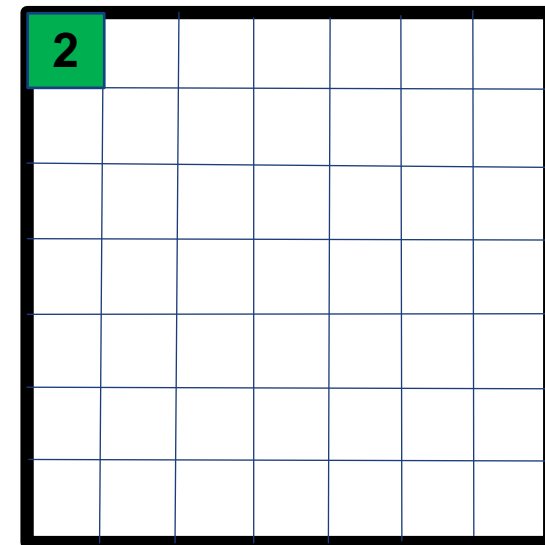
Convolutional filter



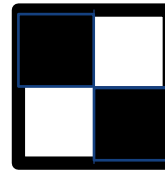
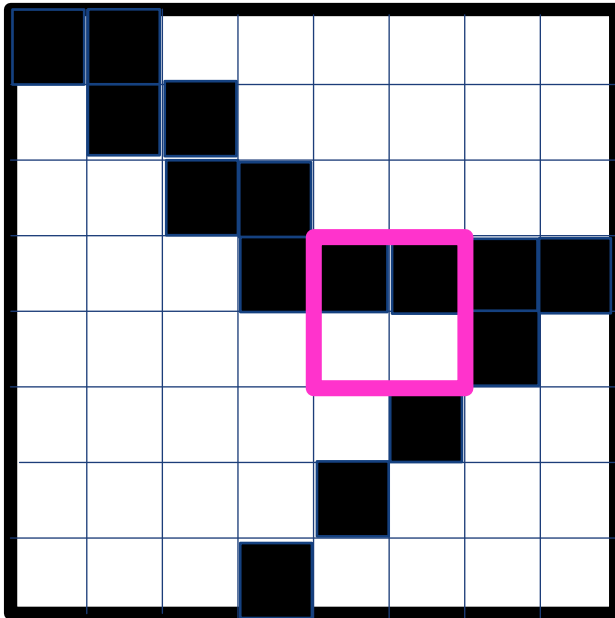
Convolutional neural networks



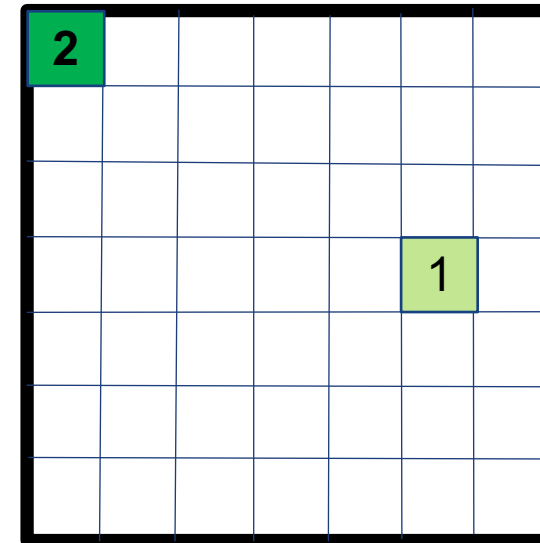
Convolutional filter



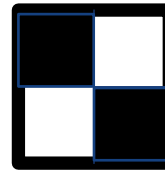
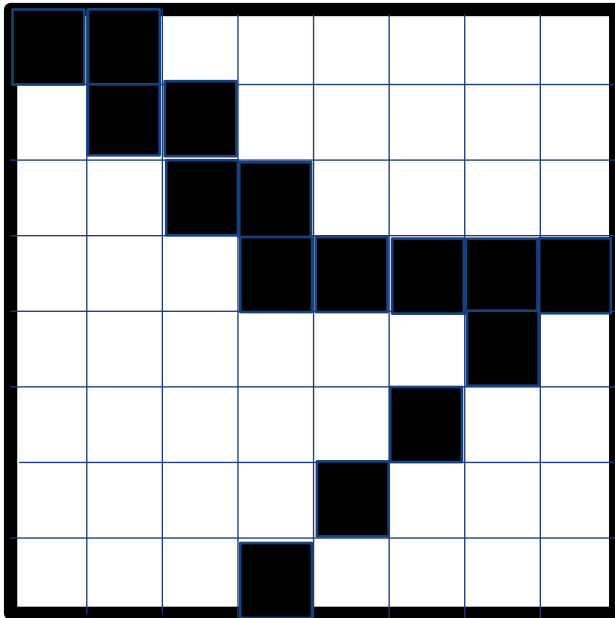
Convolutional neural networks



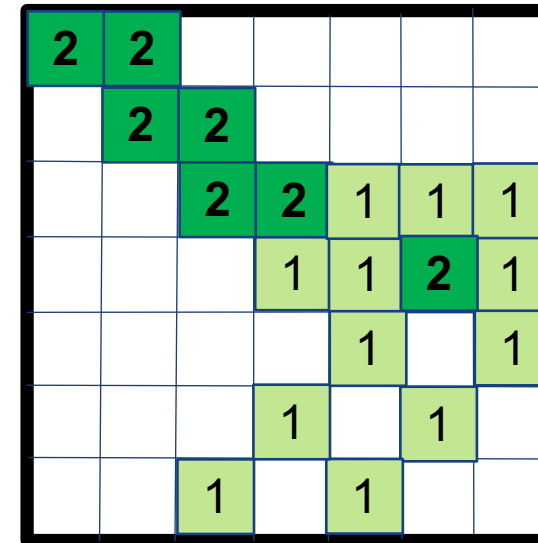
Convolutional filter



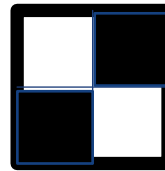
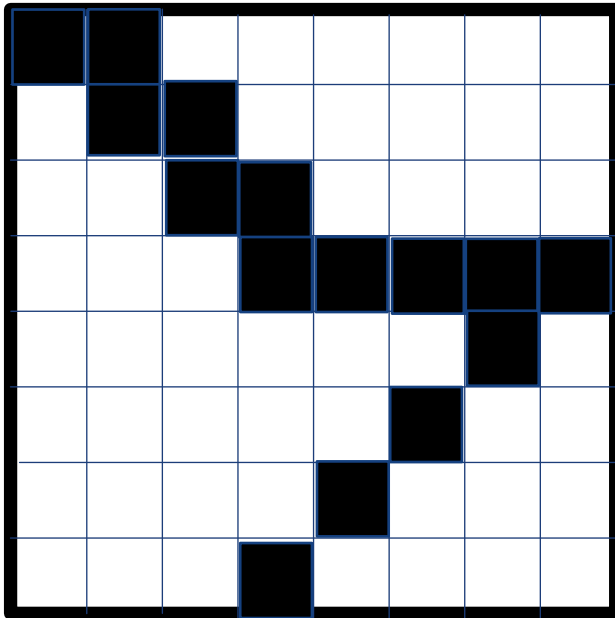
Convolutional neural networks



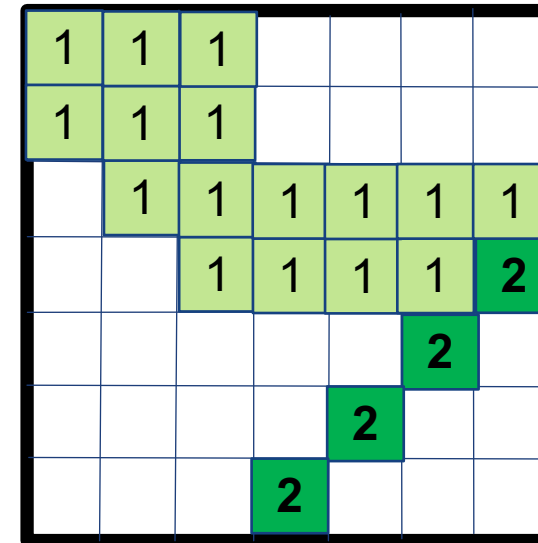
Convolutional filter



Convolutional neural networks



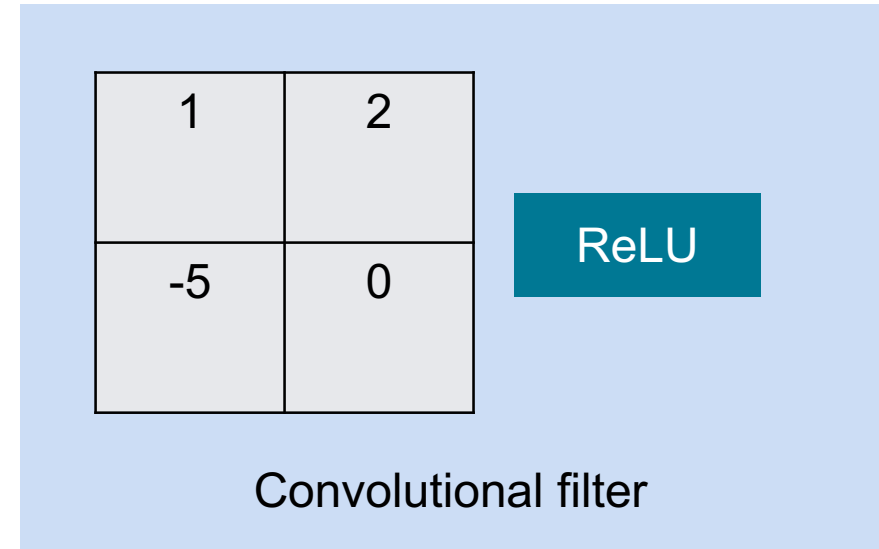
Convolutional filter



Filters

1	3	5
1	2	4
6	3	2

Input

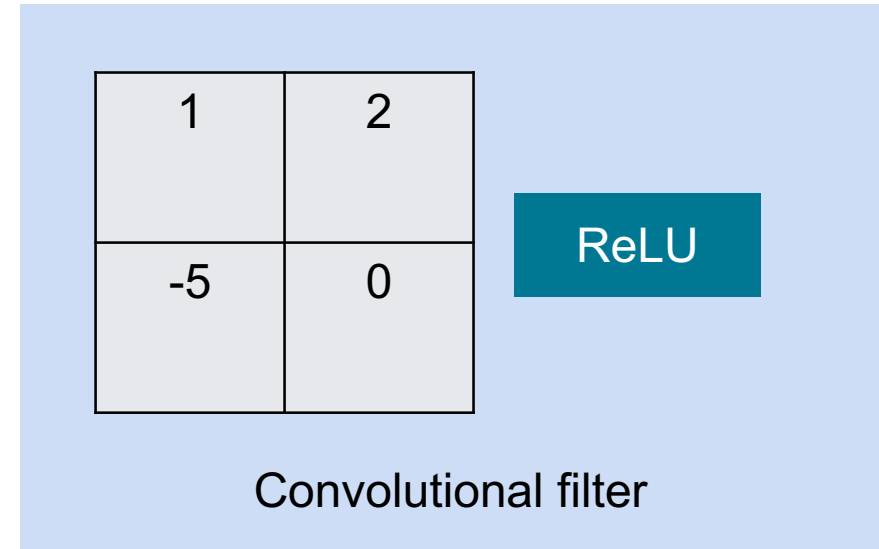


Output

Filters

1	3	5
1	2	4
6	3	2

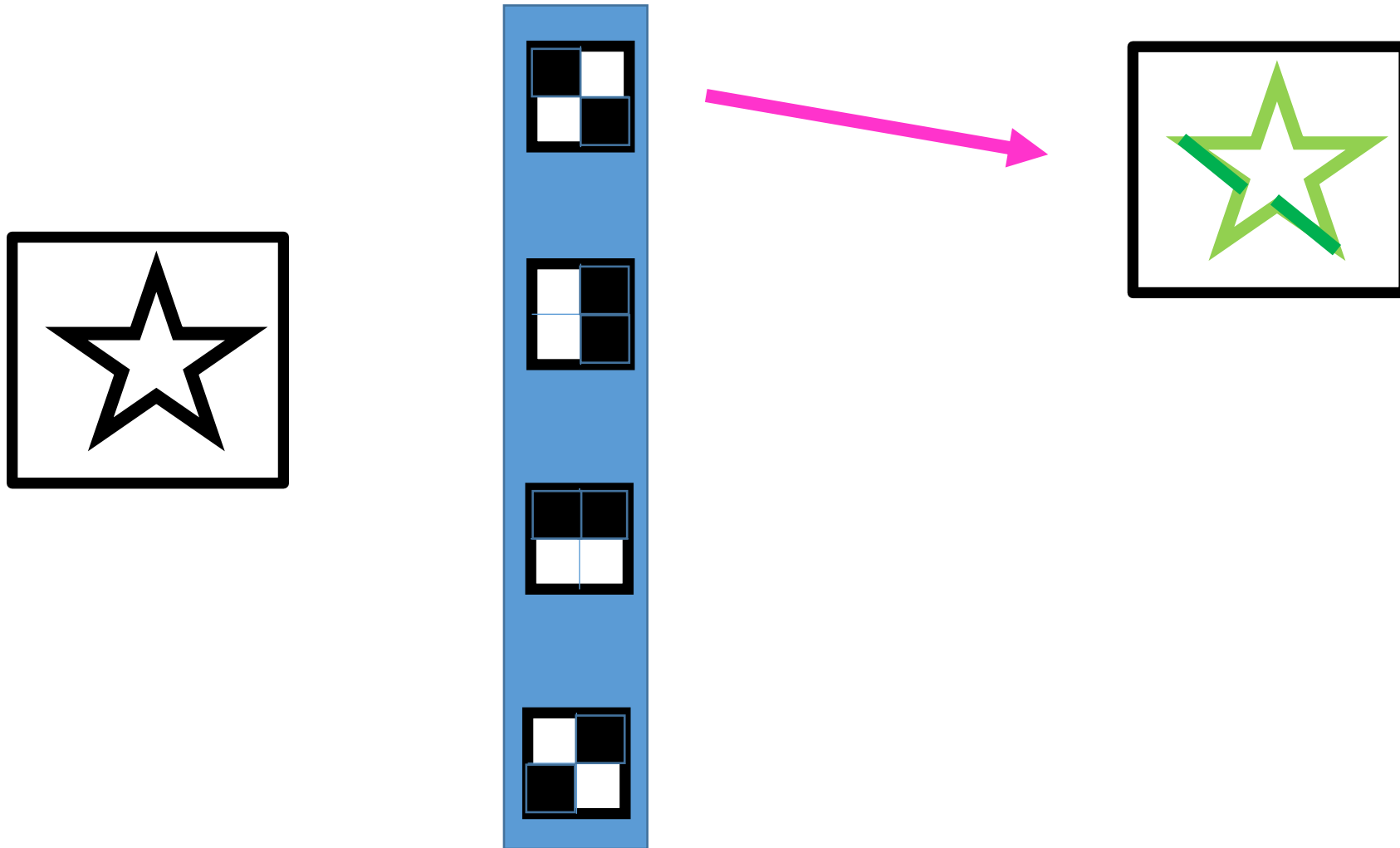
Input



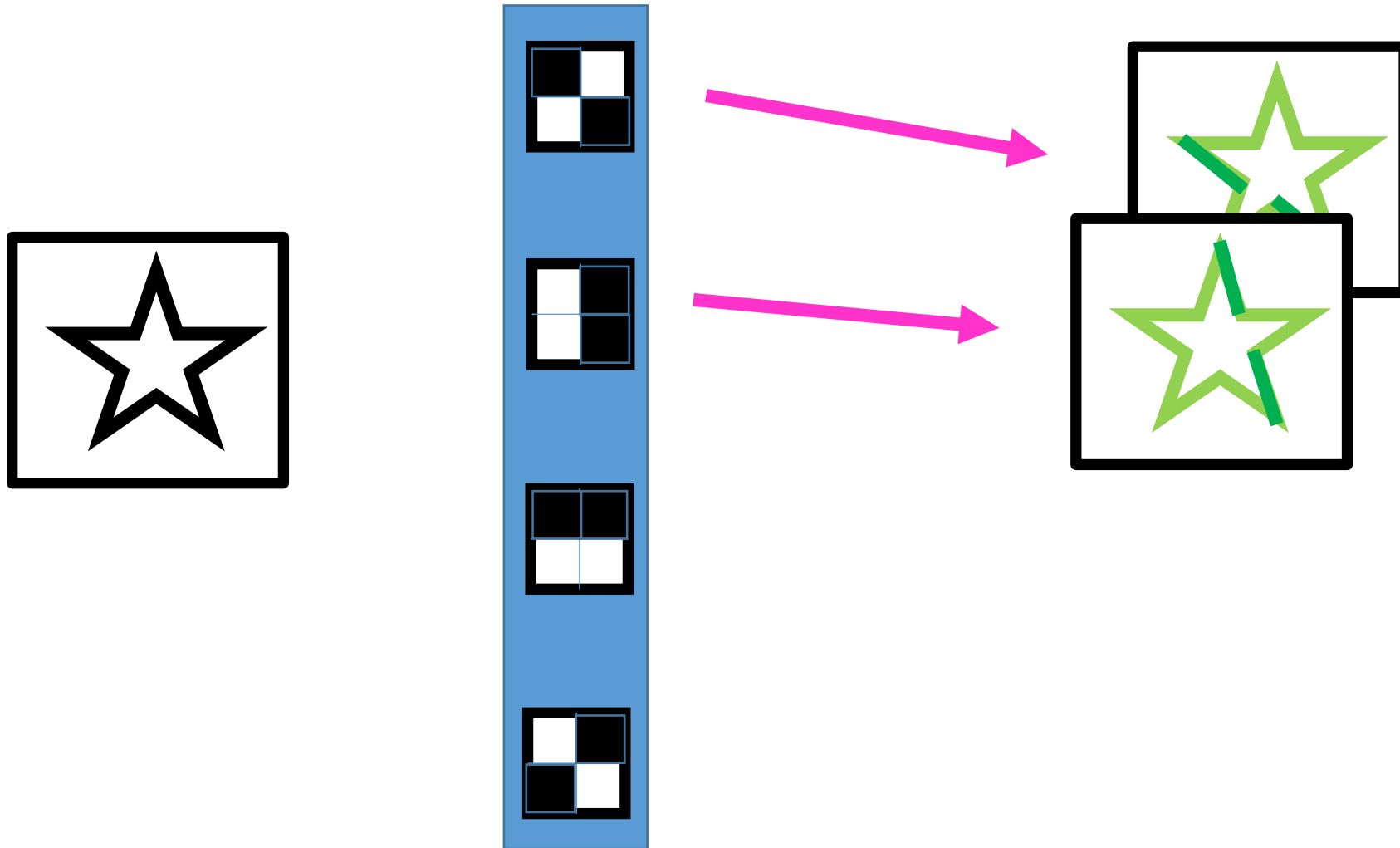
2	3
0	0

Output

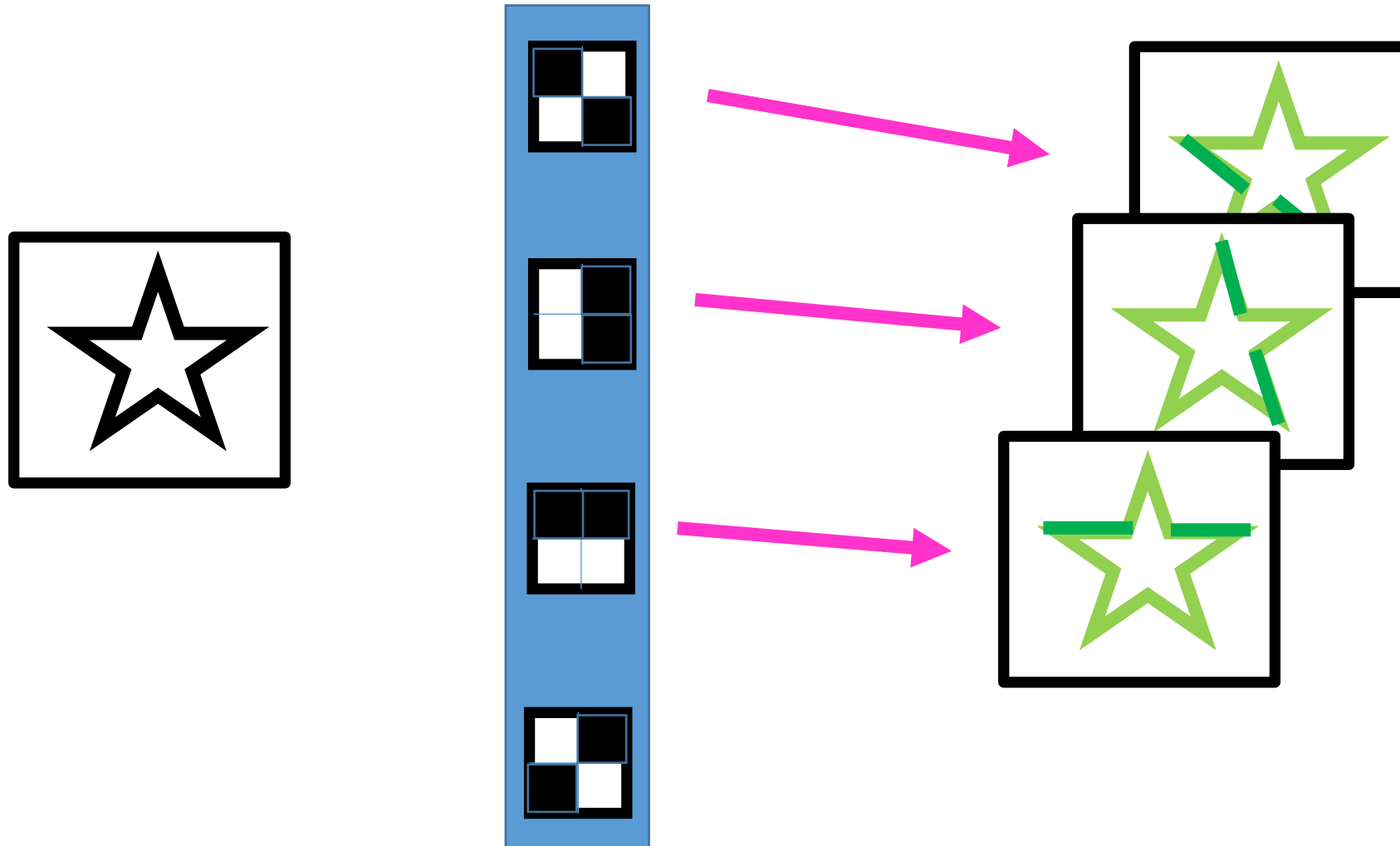
Convolutional neural networks



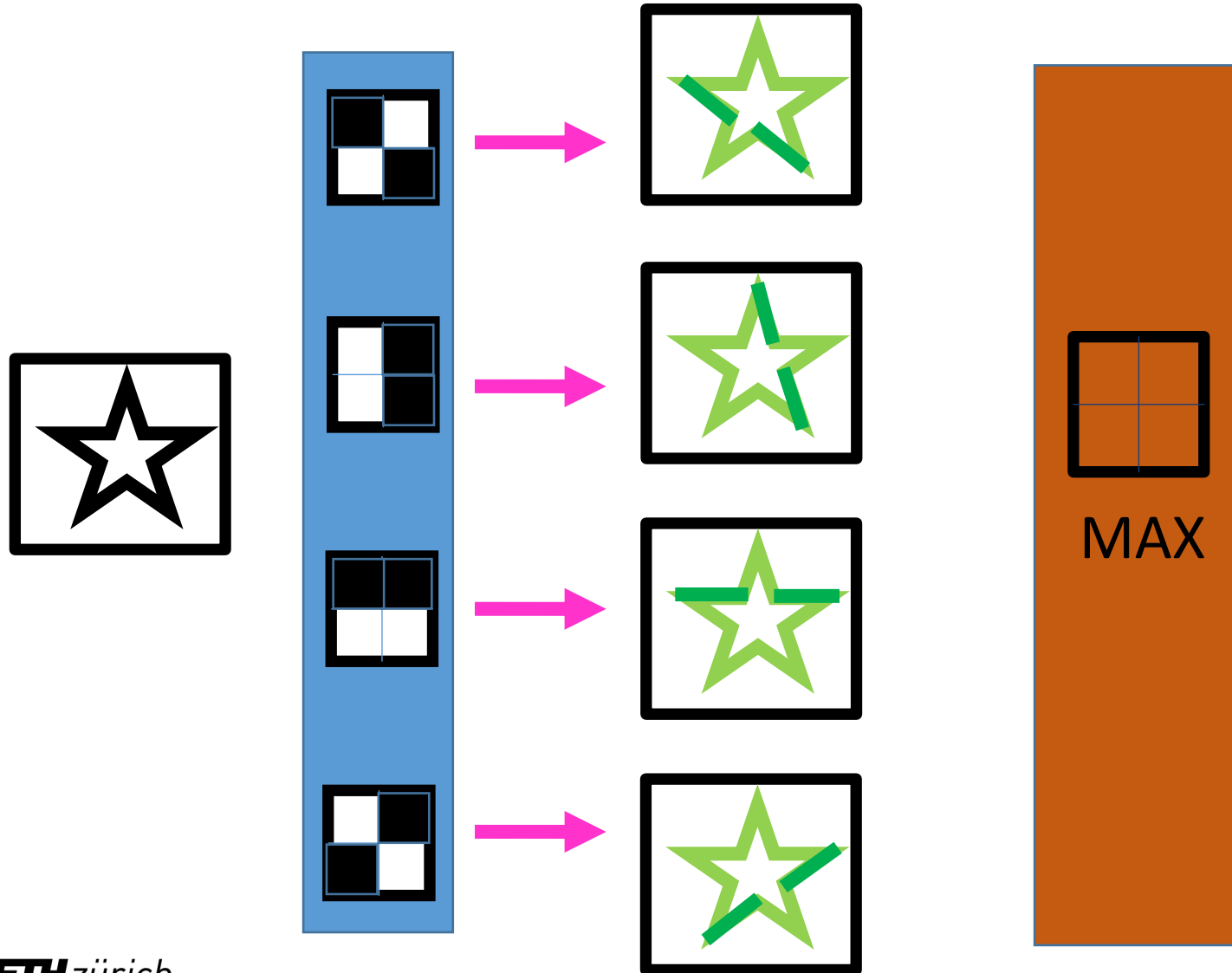
Convolutional neural networks



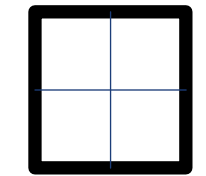
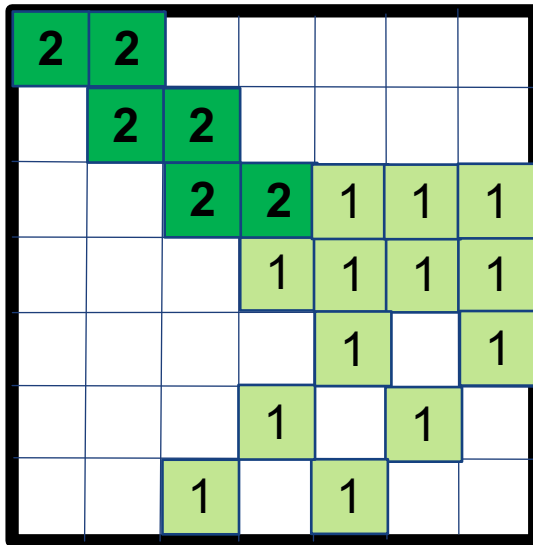
Convolutional neural networks



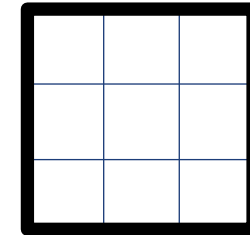
Convolutional neural networks



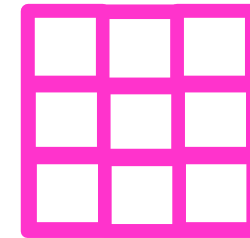
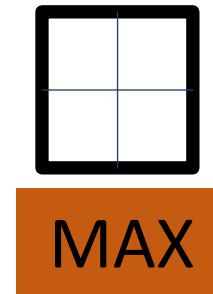
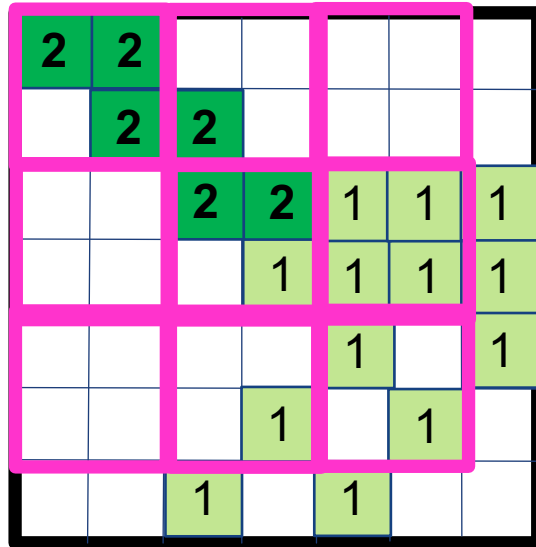
Convolutional neural networks



MAX

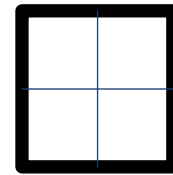


Convolutional neural networks



Convolutional neural networks

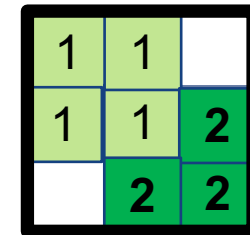
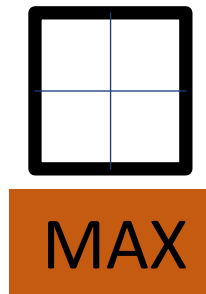
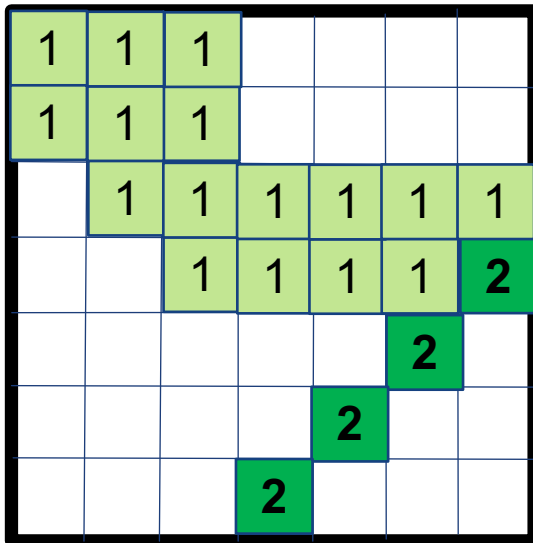
2	2					
	2	2				
		2	2	1	1	1
			1	1	1	1
				1		1
			1		1	
		1		1		



MAX

2	2	
	2	1
1	1	1

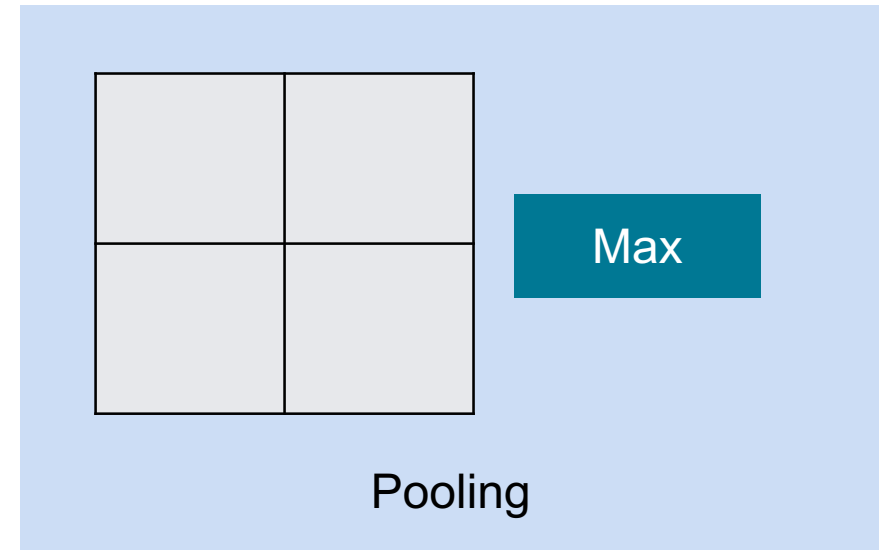
Convolutional neural networks



Pooling

1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

Input

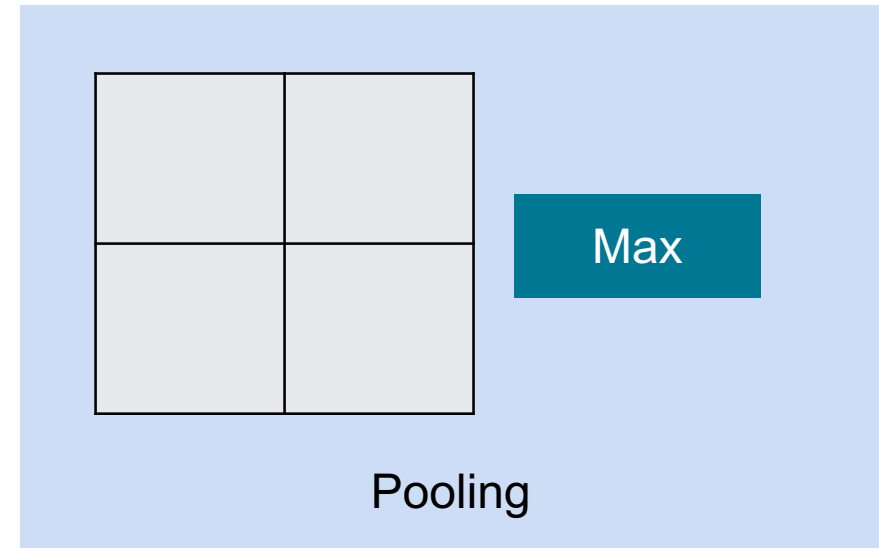


Output

Pooling

1	0	5	3	1
4	0	0	2	2
6	1	0	1	0
0	0	0	2	4
3	0	1	2	1

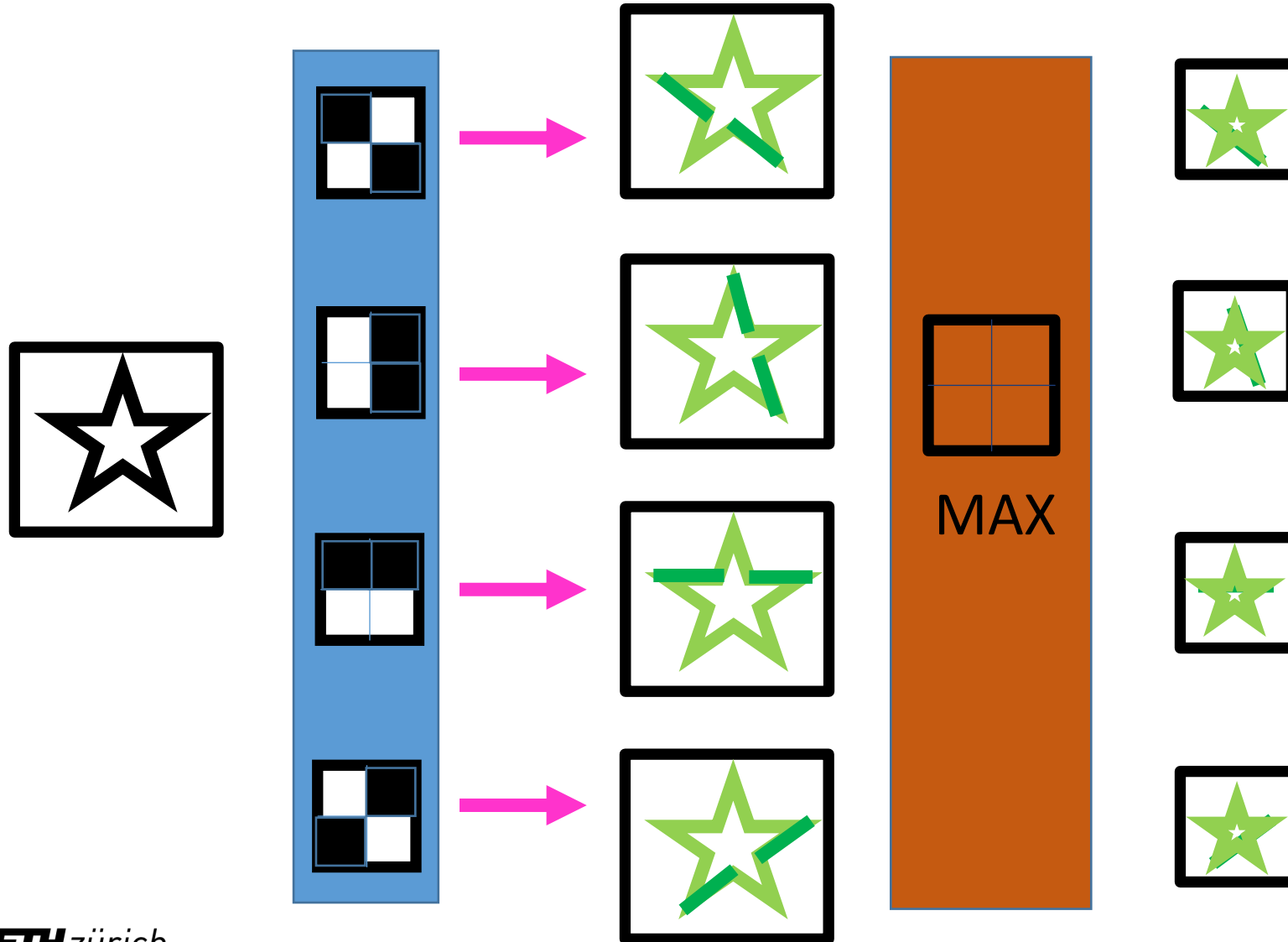
Input



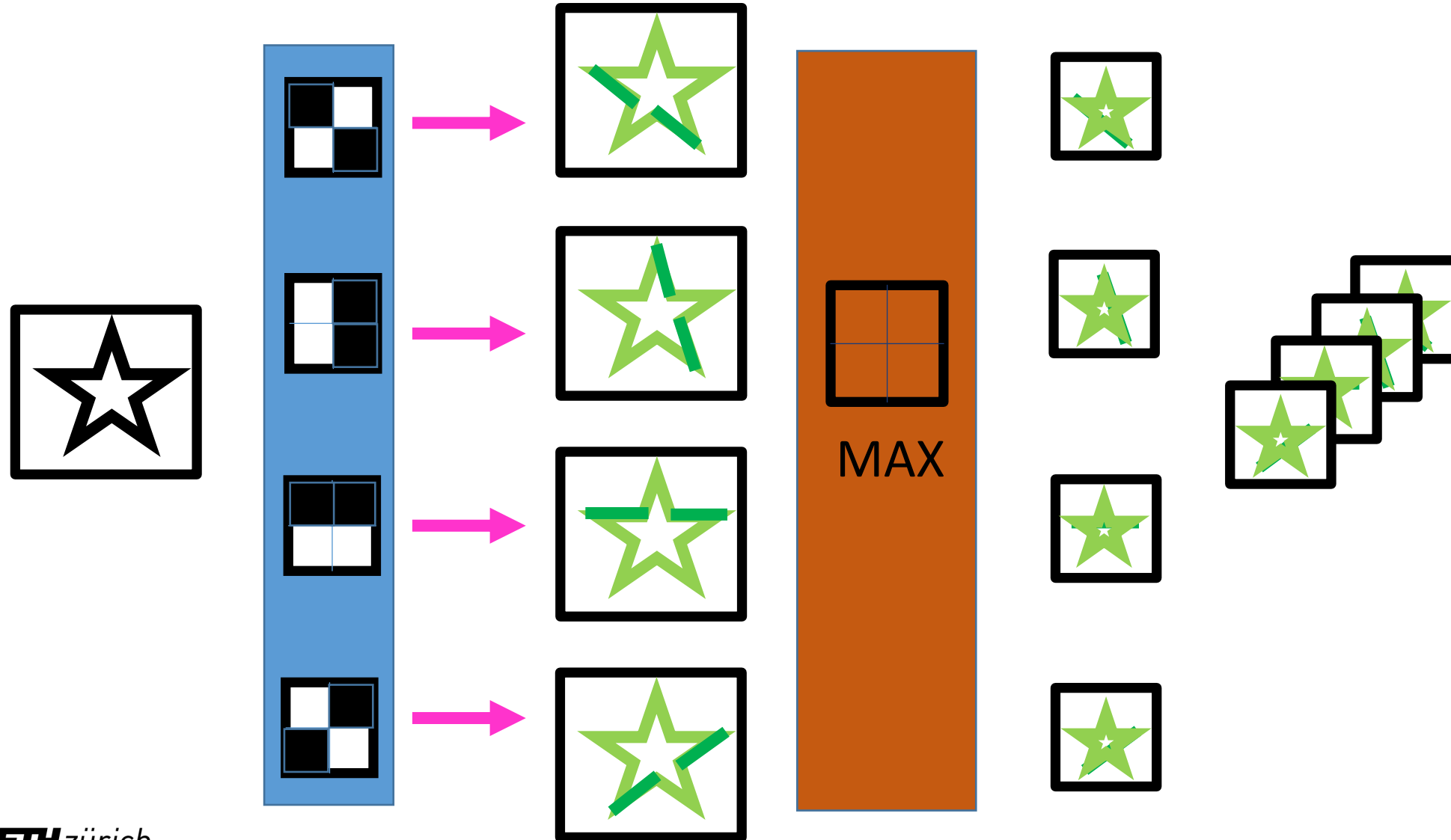
4	5
6	2

Output

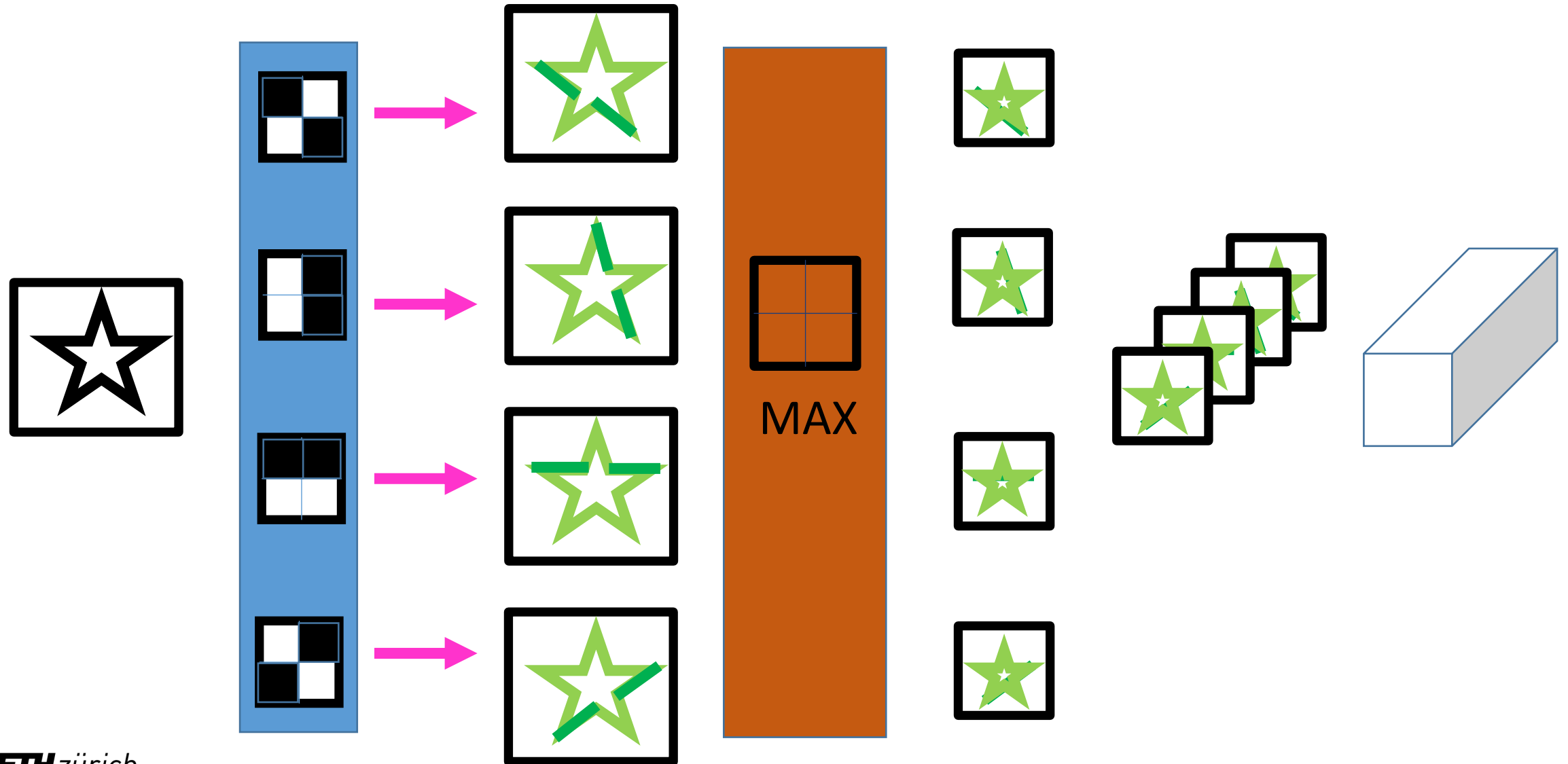
Convolutional neural networks



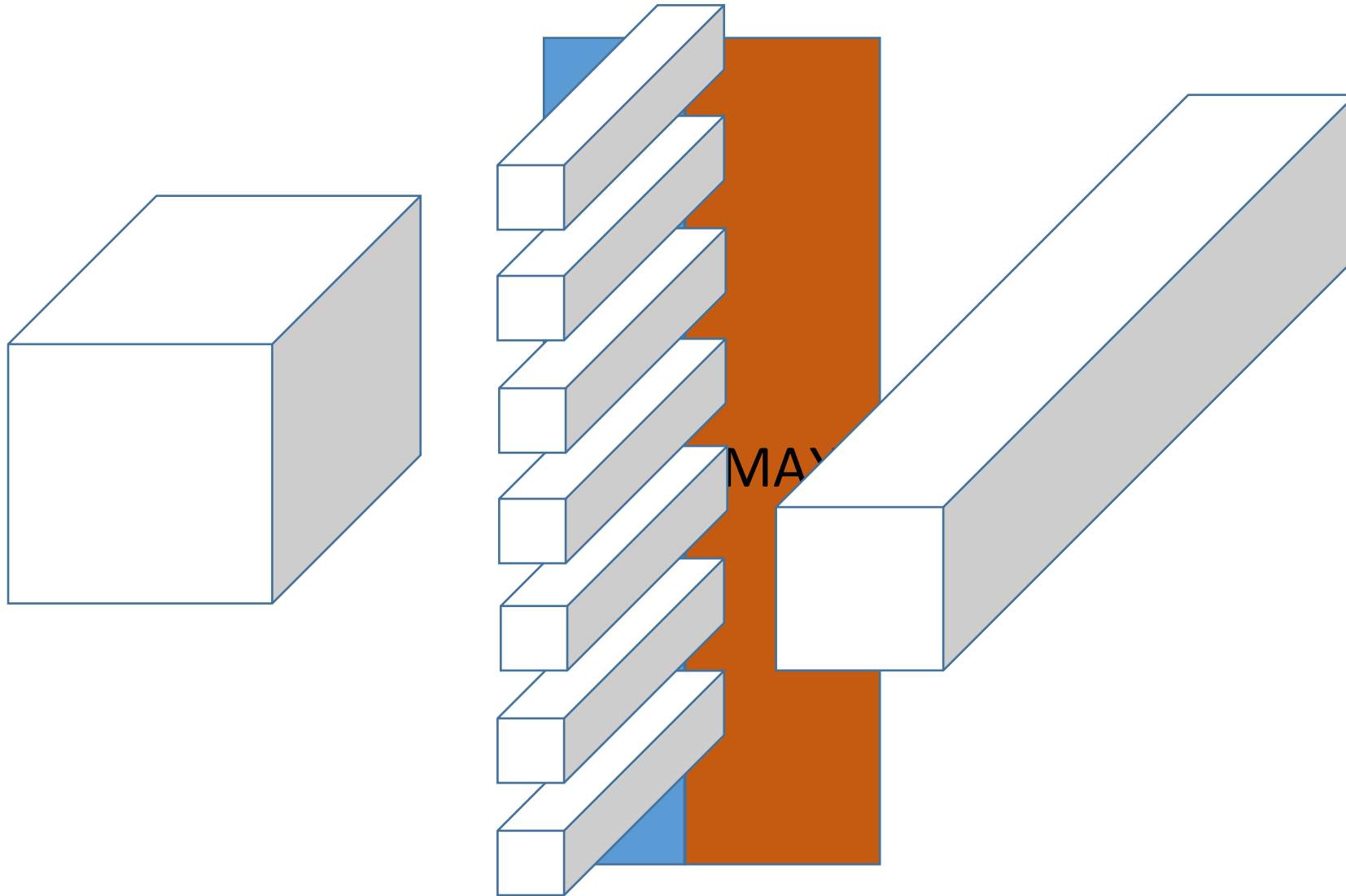
Convolutional neural networks



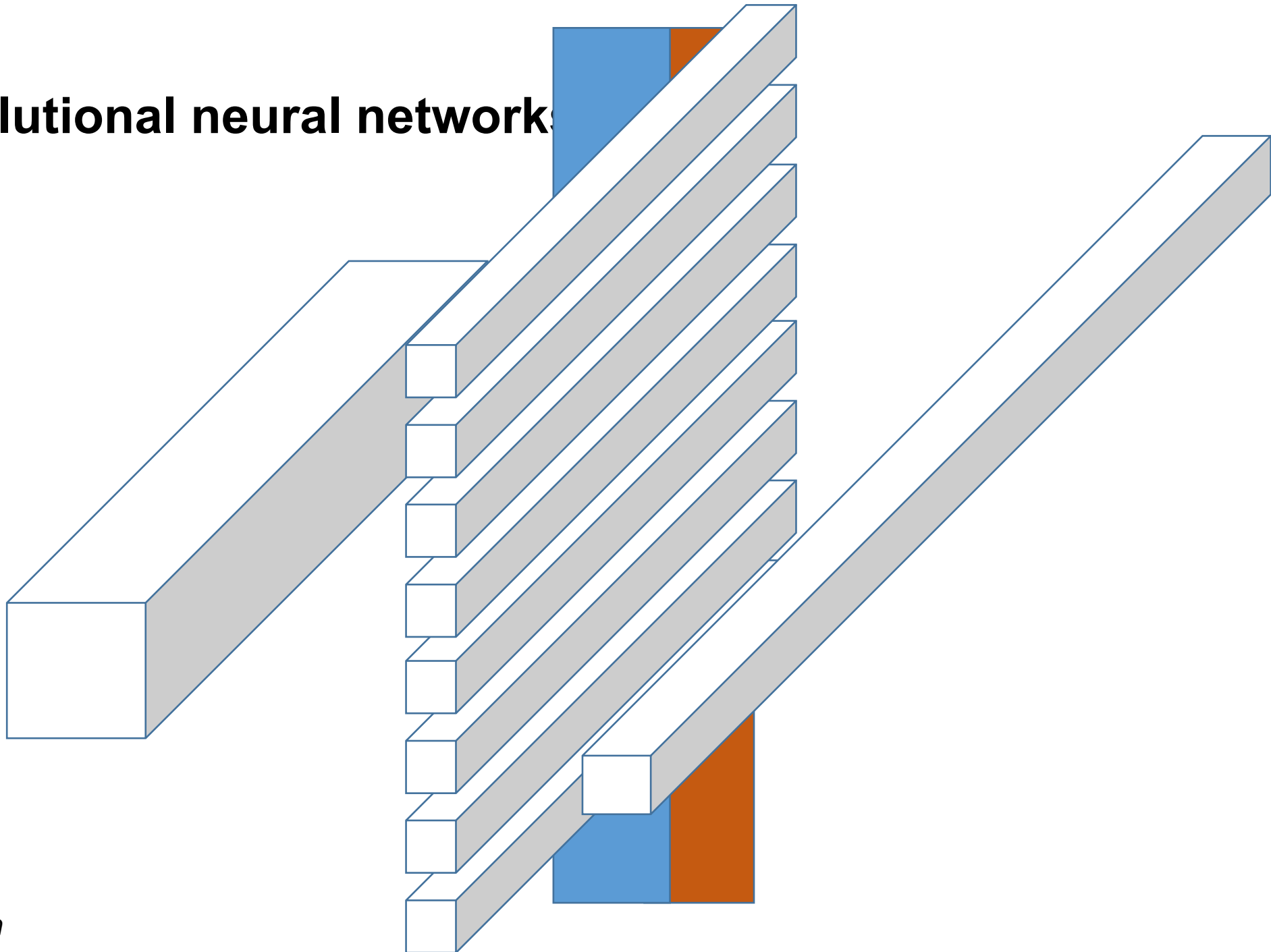
Convolutional neural networks



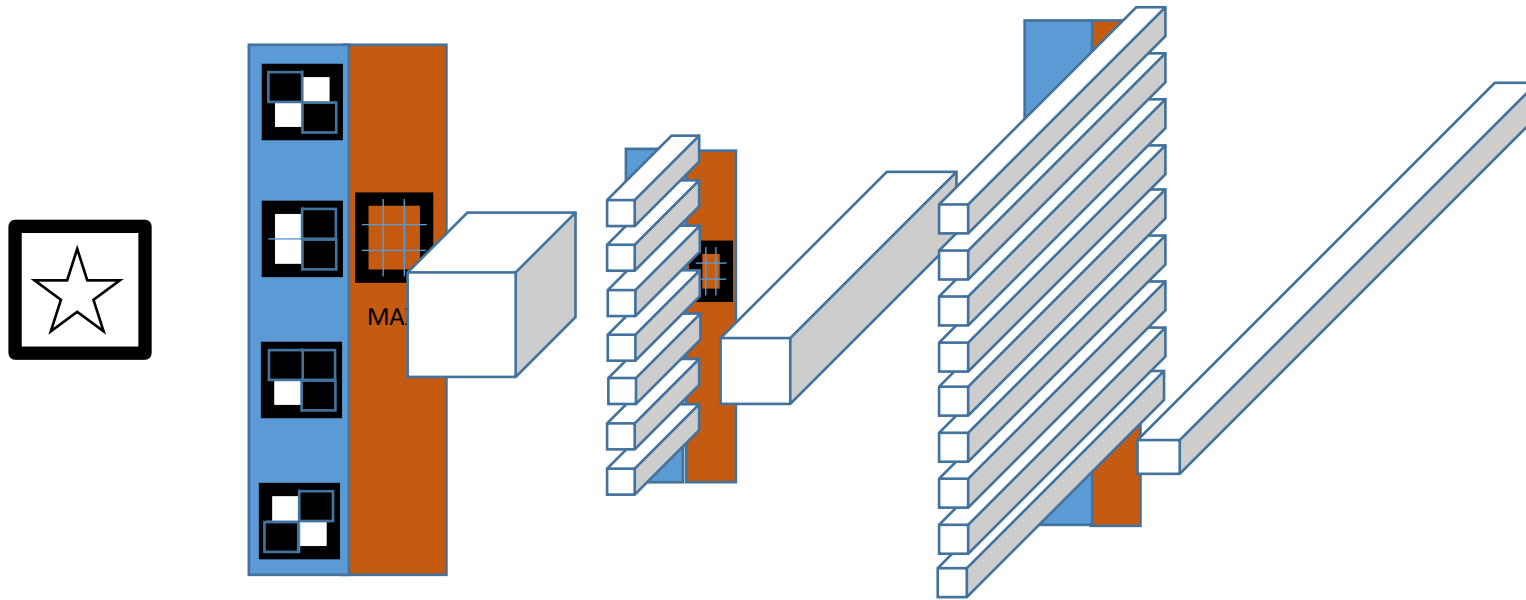
Convolutional neural networks



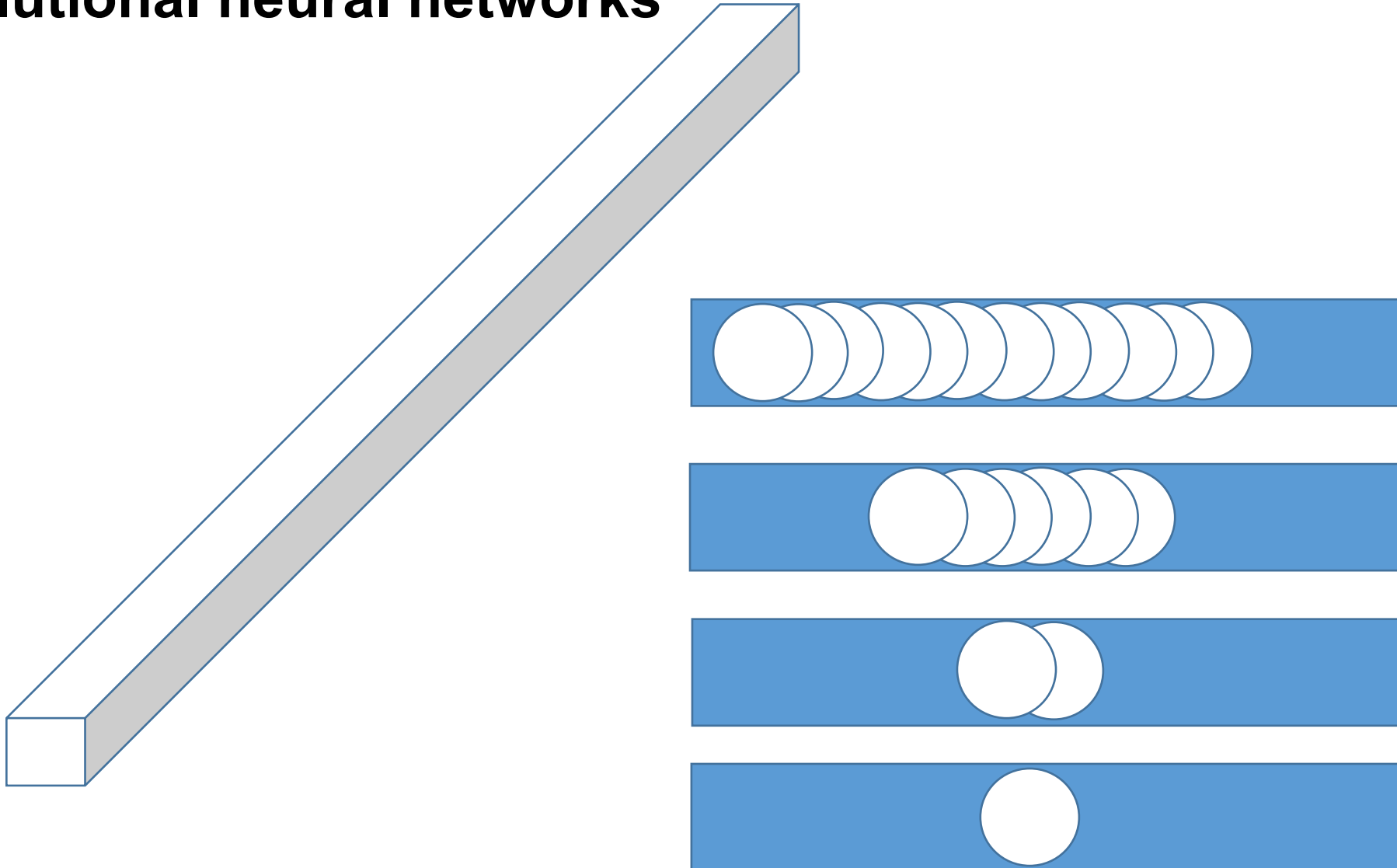
Convolutional neural networks



Convolutional neural networks



Convolutional neural networks



Convolutional neural networks

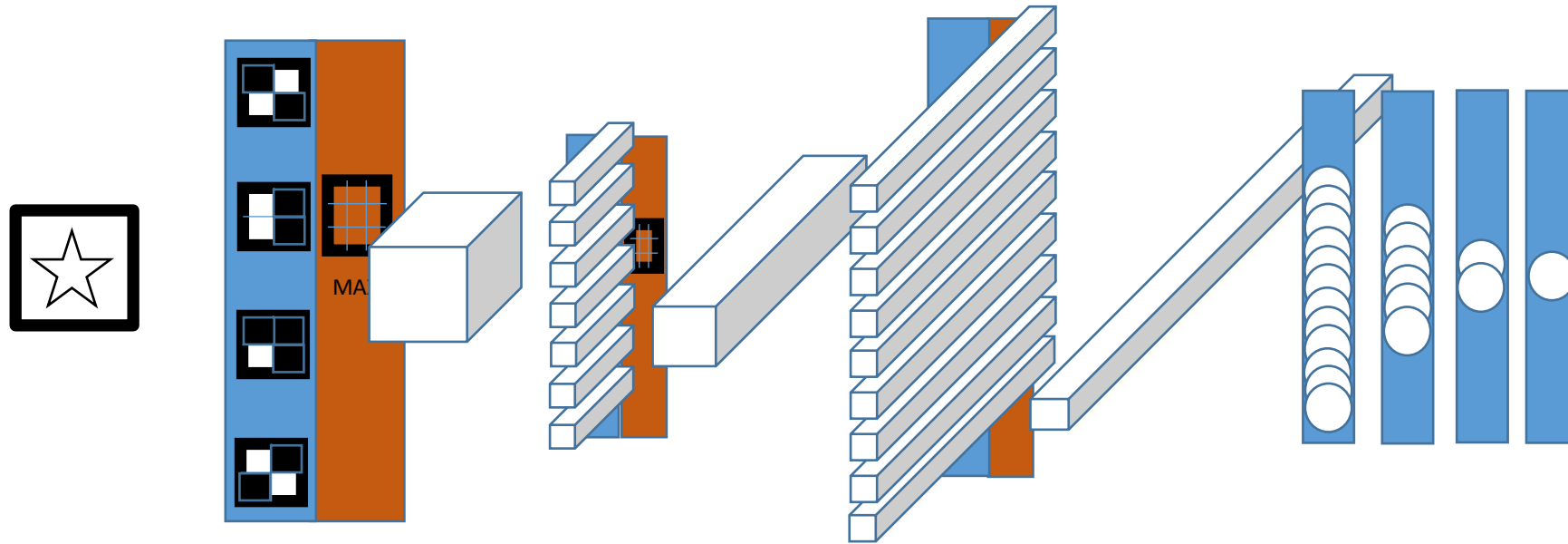
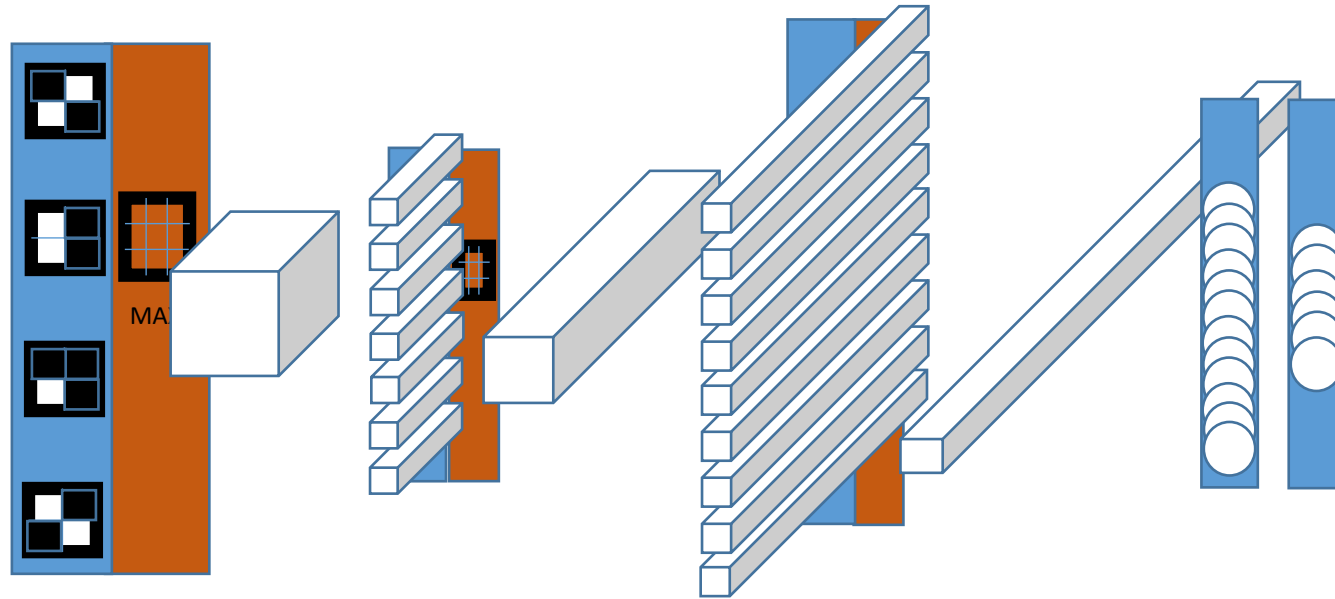
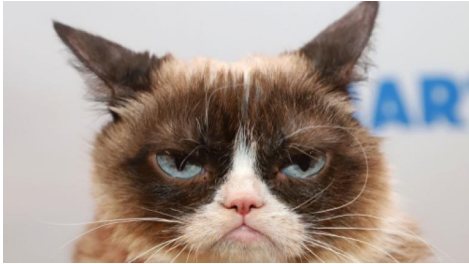


Image recognition



Softmax

